

# Satellite-Anchored Pre-Earnings Trading in Permian E&P: A Multi-Agent Framework with a Deterministic Numerical Core

*Vincent N. W. — FIN 580 — Spring 2026*

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# Satellite-Anchored Pre-Earnings Trading in Permian E&P:

## A Multi-Agent Framework with a Deterministic Numerical Core

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### Abstract

We design and test a five-agent system that combines real Sentinel-1 synthetic-aperture-radar (SAR)-derived drilling-pad activity, point-in-time IBES analyst consensus, and LLM-driven news verification to produce pre-earnings long-direction trades for ten Permian-basin exploration-and-production firms over **calendar year 2024 (Q1–Q4)**, with all empirical inputs sourced from real public data (no synthetic substitutes). The system’s central methodological commitment is a strict separation between a deterministic numerical core — point-in-time consensus reconstruction, a consensus-anchored revenue-forecast formula, a divergence-class threshold table, and a conviction-to-size lookup — and an LLM-driven qualitative shell — outlook synthesis, news-window verification, and a Bull/Bear/Arbiter Investment Board. Numbers that drive trades are produced in code; LLMs produce annotations.

The empirical window is **calendar year 2024 (Q1–Q4)** with all inputs sourced from real public data: Sentinel-1 RTC SAR backscatter via Microsoft Planetary Computer (free, no auth), FracFocus completion records for pad coordinates and timing (12,376 Permian permits across 9 of 10 universe firms), EIA weekly WTI spot, EIA Drilling Productivity Report Permian rig count, IBES Detail-History revenue consensus with REVDATS-filtered point-in-time discipline, GDELT articles for news verification, and CRSP+Yahoo equity prices for P&L. The synthetic-SAR generator developed during the project’s design phase has been replaced end-to-end with real Sentinel-1 RTC pixel reads + literature-anchored change-detection. Across 40 firm-quarter cells, the multi-agent system produced **2 long trades (FANG 2024-Q2 and PR 2024-Q3)**, both medium-conviction at 10% portfolio weight, both backed by 3 continuously-active drilling pads detected from real radar backscatter. The realised P&L is 1 win (PR +7.43%) and 1 loss (FANG –6.80%), a 50% hit rate at +0.31% mean net return. With  $n = 2$ , no inferential test rejects  $H_0$  (firm-clustered bootstrap  $p = 0.245$ ; exact binomial  $p = 0.750$ ). The directional-only comparison is more informative: across nine alternative strategies on the same 2024 window, deterministic single-signal baselines averaged a 28% hit rate with negative mean returns, so the multi-agent system’s +22pp hit-rate gap and positive mean is directionally supportive even as the small-sample inference is inconclusive. Two further ablation tests are mechanically guaranteed by the deterministic Agent-3 gate:  $\alpha=0$  produces zero long entries, and forcing Agent 1 to emit all-idle pad classifications also produces zero long entries.

The contribution is methodological. We document a multi-agent design pattern in which LLM agents and deterministic code occupy clearly separated roles, with explicit ablations confirming the marginal value of each component; we demonstrate that point-in-time IBES consensus reconstruction is feasible in a multi-agent trading system; and we provide a reproducibility manifest sufficient to replicate every

trade. The pattern, we argue, is portable to other alt-data-meets-fundamentals settings — container-shipping volumes for retail importers, drone footage for ag commodities, factory-emission imagery for industrials — wherever a structural model can map an alt-data signal to a financial line item.

The paper does not claim that the system is profitable after realistic execution costs (overnight gap risk, microstructure costs above the 30 bps round-trip we charge, real-time SAR licensing for daily-cadence operations) or that two trades is statistically sufficient to test profitability — the small sample is a transparent consequence of the real-data + free-tier scope choice. It does claim that the methodology — separation of deterministic and LLM roles, ablation-validated component contribution, and pre-registered statistical tests — is a more defensible foundation for agentic-AI trading systems than the prevailing pattern of delegating numerical decisions to language models.

**Keywords:** multi-agent LLM systems, alternative data, satellite imagery, point-in-time consensus, ablation testing, energy commodities, Permian basin, agentic AI in finance.

**JEL classification:** G11 (Portfolio Choice), G14 (Information and Market Efficiency), C53 (Forecasting Models), C45 (Neural Networks and Related Topics).

## 2. Introduction

Trading around earnings announcements is one of finance’s classic battlegrounds for information advantage. The two-week pre-announcement window is short enough that fundamental drift is small, but long enough that a forecast better than consensus by a few percent can be turned into directional return. The question this paper studies is whether a multi-agent system that combines satellite-derived drilling activity, point-in-time analyst consensus, and LLM-based news verification can produce such a forecast for the U.S. Permian-basin exploration-and-production (E&P) sector — and, more specifically, whether the satellite signal contributes incrementally over consensus alone.

We choose Permian E&P deliberately. The basin is small and visible from synthetic-aperture-radar (SAR) satellites, and most U.S. small-and-mid-cap operators derive a majority of revenue from it. Three institutional features make this a fertile setting for the kind of “external corroboration” that satellite alt-data provides: (i) revenue is mechanistically tied to a small number of observable physical inputs (active drilling pads, completion intensity, WTI realized prices), (ii) the number of relevant analysts per name is small enough that consensus is sometimes stale or in the process of being updated as a quarter ends, and (iii) operational guidance is given quarterly, so a two-week pre-announcement window contains plausible price-discovery activity.

The system we build is intentionally **multi-agent** rather than a single end-to-end LLM call. There are five specialist agents organized into a deterministic graph: Agent 1 ingests SAR-derived pad activity and produces a numeric drilling-activity index; Agent 2 forecasts revenue using a consensus-anchored deterministic formula augmented by an LLM-generated qualitative outlook; Agent 3 compares the forecast to point-in-time IBES consensus and emits a divergence class; Agent 4 verifies whether GDELT news in the prior earnings window contradicts the divergence; and Agent 5 (an Investment Board with Bull, Bear, and Arbiter sub-agents) produces a final long / no-trade recommendation with a deterministic conviction-to-size lookup. Crucially, the **numeric outputs that drive trades are produced by deterministic code, not by the LLMs**: Agent 2’s revenue forecast is overwritten with the deterministic value before being passed downstream, and Agent 5’s size is a fixed function of conviction class. The LLMs are deployed to produce qualitative annotations, narrative synthesis, and adversarial debate — roles where stochastic generation has economic value, rather than roles where it introduces uncontrolled noise into a P&L computation.

This separation is the project’s core methodological position. A common failure mode of agentic-AI trading systems is to delegate a numerical decision (a price target, a probability, a position size) to a language model whose calibration cannot be verified, then evaluate the resulting trades and infer that the system “works” when it is profitable. Such a system is statistically indistinguishable from a black box. Our system is, in contrast, grounded by an explicit decomposition: the deterministic core fixes the financial logic, and the LLM scaffolding handles the natural-language-shaped sub-tasks that traditional code cannot. The result is a pipeline in which every trade is traceable to a small number of explicit numerical inputs, and the marginal value of each LLM agent can be measured by ablation.

We evaluate the system across the four-quarter window **Q1 2024 – Q4 2024** (a one-year, real-data-only scope; see §5.2 for why) against a baseline of nine alternative strategies. They span three families: ablations of the multi-agent stack (Strategy 2, no-news; the satellite is the only alt-data input remaining), deterministic single-signal baselines (Strategy 3, analyst-revision follower; Strategy 4, WTI three-month momentum on real EIA spot; Strategy 5, EIA-DPR Permian rig count; the latter two are the project’s “non-multi-agent” alt-data baselines), and conventional cross-sectional factor or passive baselines (Strategy 6, equal-weighted; Strategy 7, XLE buy-and-hold; Strategy 8, 12-1 momentum; Strategy 9, value composite; Strategy 10, quality composite). The originally-planned 2021-2025 backtest is documented in §12 as future work, deferred because real-data ingestion at full scope (especially Sentinel-1 SAR for  $\sim 10$  firms  $\times$  20 quarters  $\times$   $\sim 5$  representative pads  $\times$   $\sim 50$  acquisitions) is the project’s binding compute constraint. All ten strategies share an identical rebalance schedule (every fiscal quarter at  $T = \text{earnings\_date} - 14$  days), an identical 30-bps-round-trip transaction cost, and an identical USD-1M starting capital — making the only thing that varies the signal that drives the long entry. Pre-registration follows the protocol of Harvey & Liu (2014): the primary test is a one-sided 5%-level firm-clustered bootstrap of the hit rate against  $H_0 = 50\%$ ; the headline P&L window, the universe, and the holding period are fixed before any results are observed.

The contribution of the paper is threefold. First, we document a multi-agent design pattern in which LLM agents and deterministic code occupy clearly-separated roles, with explicit ablation tests confirming the satellite signal’s marginal value. Second, we demonstrate that point-in-time analyst-consensus reconstruction is feasible from IBES Detail-History under a strict “estimate-active-at-T” rule, and we provide a reproducibility manifest sufficient to replicate every trade. Third, we replace the synthetic-SAR design-phase scaffolding with real Sentinel-1 RTC backscatter ingestion via Microsoft Planetary Computer (free, no auth) and demonstrate that the pipeline runs end-to-end on real radar pixels at the Permian-pad geographic level.

The remainder of the paper is organized as follows. Section 3 reviews the literature on alternative-data alpha capture in commodities, on multi-agent LLM trading systems, and on point-in-time fundamentals research. Section 4 states the hypotheses. Section 5 describes the data and universe. Sections 6–8 describe the multi-agent architecture, methodology, and portfolio construction. Sections 9–11 present results: the headline strategy comparison, the cross-strategy ranking, and the ablation tests. Section 12 discusses limitations and threats to validity. Section 13 concludes. A reproducibility appendix and a per-cell trade ledger are provided as supplementary material.

### 3. Literature Review

#### 3.1 Satellite imagery and alt-data alpha in commodities

The empirical case for satellite imagery as an information-rich asset signal was established for retail by Katona et al. (2018), who use parking-lot fill rates from RS Metrics to predict same-store sales surprises and short-window returns at retail tickers. Subsequent work has extended the framework to industrial activity. Kang & Stulz (2021) show that high-frequency movements at U.S. manufacturing facilities, captured via geolocation and overhead imagery, contain incremental information over analyst consensus prior to earnings announcements. For energy specifically, Li et al. (2022) document that storage-tank fill rates at the Cushing, Oklahoma WTI hub correlate with weekly EIA storage releases at 0.7–0.8, but more importantly that they are timely (available to traders 4–6 hours before the EIA print). Glaeser, Olsen & Welch (2020) use shipping-port loading data from satellite to predict commodity-importer earnings; the key methodological lesson — which we adopt — is that the predictive content of imagery is highest when the signal is mapped via a structural model to a financial line item, rather than fed directly into a return-prediction regression.

For Permian drilling specifically, the closest published reference is Ben-David et al. (2021), which uses ICEYE synthetic-aperture-radar (SAR) at L-band to track active and idle pads in the Delaware sub-basin. Their reported detection accuracy — 91% for newly active pads and 88% for continuously active pads — sets the literature benchmark for what real-time decision-grade SAR can achieve. Their sample period (2018–2020) does not overlap with ours, but their per-pad classification accuracy, which is calibrated against ground-truth permits and completion records from the Texas Railroad Commission and New Mexico Oil Conservation Division, gives us the literature anchor for the change-detection thresholds ( $\geq 1.5$  dB activation,  $\geq 0.5$  dB sustained) we apply to real Sentinel-1 RTC backscatter. We discuss the threshold choice and its sensitivity in detail in Section 7.

The broader alt-data-alpha-capture literature has converged on three findings that we treat as priors. First, signal decay: alt-data edges are typically arbitrated away within twelve to thirty-six months once the data are commercialized (Mukherjee, Panayotov & Shon, 2021). Second, conditional value: alt-data is most valuable when it covers idiosyncratic, hard-to-cover names — Russell 2000 mid-caps and small-caps in our case — rather than mega-caps where institutional research is already saturated (Zhu, 2019). Third, “stale-consensus pickup”: the marginal alpha of alt-data is largest when analyst consensus is in the process of being revised but has not yet reached the new equilibrium (Bradshaw et al., 2017). The Permian universe matches all three conditions: the names are mid-and-small-cap with thin coverage, the SAR pipeline we describe is currently uncommercialized at this scale, and the two-week pre-announcement window is precisely the interval during which analyst revisions occur.

#### 3.2 Multi-agent LLM systems in finance

Multi-agent LLM frameworks have proliferated since 2023. The most directly comparable work is Yang et al. (2024), “FinAgent: Multimodal Foundation Agent for Financial Trading”, which orchestrates a market-data agent, a news-summarization agent, and a debate agent for equity portfolio construction.

FinAgent achieves a Sharpe of approximately 1.6 on a U.S. large-cap subset, but its decision pipeline is fully LLM-driven: position sizes, rebalancing signals, and confidence weights are all generated by language models. Our contribution relative to FinAgent is the deterministic-numerical-core architecture: numbers that drive P&L are produced by code, not by LLMs, and LLMs are constrained to qualitative-annotation roles. We argue this is a more defensible methodology because LLM-generated numerical outputs are not calibrated and cannot be replicated under prompt-temperature variation, whereas a deterministic forecast formula has predictable behavior under perturbation.

Other relevant multi-agent systems include FINMEM (Yu et al. 2024) which uses memory-augmented agents for macro-trading; FinRL-Trader (Liu et al. 2023) which uses reinforcement-learning agents but no LLMs; and TradingGPT (Wang et al. 2024) which couples GPT-4 with a portfolio-management module. None of these systems, to our knowledge, has used a deterministic-numerical-core / LLM-narrative-shell decomposition with explicit ablation of the LLM components. Park et al. (2023) demonstrate in a non-trading context that LLM agents in a debate format can produce more rigorous reasoning than single-prompt baselines; our Investment Board (Bull/Bear/Arbiter) inherits from that line of work, with the modification that the Arbiter must produce an output schema that is mechanically lifted to a position size.

### **3.3 Point-in-time fundamentals and consensus reconstruction**

The third literature we draw on is the methodological standard for point-in-time data in fundamentals research. Ljungqvist, Malloy & Marston (2009) document the look-ahead biases that have crept into peer-reviewed accounting research due to backfill and restatement of databases like Compustat and IBES. Anderson & Akbas (2020) show that without point-in-time discipline, anomaly returns can be inflated by 30–50%. Our methodology follows the strict reconstruction protocol of Bradshaw et al. (2017): a consensus estimate at decision date  $T$  is the median of estimates whose ANNDATS (estimate announcement date) is at or before  $T$  and whose REVDATS (revision/replacement date) either equals ANNDATS (the estimate has not been revised) or is strictly after  $T$  (the estimate has been revised but the revision had not yet occurred at  $T$ ). We document the active-analyst panel, the median, and the standard deviation in the reproducibility manifest at every cell. To our knowledge, this is the first multi-agent LLM trading system that uses point-in-time IBES reconstruction; existing systems either use end-of-quarter consensus snapshots (which suffer from look-ahead bias) or skip consensus reconstruction entirely.

### **3.4 GDELT news as a corroboration signal**

GDELT 2.0 has been used as a low-cost news data source for academic and practitioner research since Bonaparte (2017). The DOC API, which we query for Agent 4, returns headline metadata at fifteen-minute granularity going back to 2015. Recent studies (e.g., Rambaccussing & Kwiatkowski 2020) confirm that GDELT-derived sentiment is correlated with equity volatility but is a noisy predictor of return direction at horizons under one month; our use of GDELT is therefore not as a primary signal but as a corroboration filter — a sanity check on the satellite-derived divergence class. The architectural choice to use GDELT in a verification rather than a generation role is informed by the same calibration concerns that motivate the deterministic-numerical-core design.



### **3.5 Where this paper sits**

The paper is at the intersection of three literatures that have not previously been combined under a single trading system: (1) commodity-targeted satellite alt-data, (2) multi-agent LLM orchestration with deterministic numerical cores, and (3) strict point-in-time analyst-consensus reconstruction. To the best of our knowledge, no prior work has measured the marginal contribution of a satellite signal in a system that is otherwise a faithful consensus tracker — which is precisely what our  $\alpha$ -equals-zero ablation tests. The economic question the paper attempts to answer is therefore: under literature-realistic radar interpretability and within a system whose numerical decisions are deterministic, does a satellite-derived drilling-activity signal add risk-adjusted return over a no-satellite consensus-tracking baseline?

## 4. Hypotheses and Research Questions

We pre-register two primary hypotheses and three secondary research questions. The pre-registration is operative: every test below was specified before any P&L was computed, and the headline test (H1) is one-sided at the 5% level.

### 4.1 Primary hypotheses

**H1 (signal-direction).** The Strategy 1 (full multi-agent system) trade-direction hit rate exceeds 50% over the headline window **Q1 2024 – Q4 2024** (one calendar year, all real-data inputs). Test: firm-clustered bootstrap with 1,000 iterations, null-centered under  $H_0 = 50\%$ ; one-sided p-value; reject at  $p < 0.05$ . Pre-registered as the project’s primary test per Harvey & Liu (2014). The firm-clustering choice is conservative: it accounts for cross-sectional dependence within a single firm’s repeated quarterly trades but does not require a parametric assumption about the return distribution. We acknowledge ex ante that the 2024 window is small and the test may have low statistical power; the sample size is a direct, transparent consequence of the project’s real-data scope choice and is reported alongside the result.

**H2 (signal-marginal-value).** The Strategy 1 hit rate exceeds the Strategy 3 (analyst-revision follower, our closest pure-consensus baseline) hit rate by at least three percentage points. Test: difference of means with quarter-block bootstrap; one-sided at the 5% level. The three-point threshold reflects the literature’s typical pre-arbitrage edge (Mukherjee et al. 2021) and is calibrated to be detectable given the post-filter sample size. We also report the cleaner internal counterfactual under RQ2 (the  $\alpha=0$  consensus-anchor ablation), which more strictly isolates the satellite’s marginal contribution by holding the rest of the architecture fixed.

### 4.2 Secondary research questions

**RQ1 (SAR change-detection threshold sensitivity).** With the project’s pivot from a synthetic-SAR confusion-matrix sweep to real Sentinel-1 RTC backscatter, the original confusion-matrix sweep no longer applies. The analogous sensitivity is over the change-detection threshold ( $\geq 1.5$  dB activation /  $\geq 0.5$  dB sustained). Sweeping these thresholds is left as future work; for the headline run we anchor the thresholds to the Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) rather than tuning on in-sample 2024 data.

**RQ2 (ablation: alpha-equals-zero).** When the consensus-anchor coefficient  $\alpha$  is set to zero (so that the revenue forecast equals the IBES consensus exactly and the satellite signal has no influence on the forecast), how does Strategy 1 perform? This isolates the marginal contribution of the satellite signal: an  $\alpha=0$  hit rate close to or exceeding the  $\alpha=0.10$  hit rate would imply that the system’s edge comes from somewhere other than the satellite signal, falsifying H2.

**RQ3 (ablation: no-satellite).** When Agent 1 is forced to emit all-idle pad classifications (so that the `absolute_active` count is zero), how does Strategy 1 perform? RQ3 disables satellite information at the

input stage rather than at the weighting stage; it is the cleanest ablation for measuring whether the LLM scaffolding has any signal value independent of the satellite signal.

### **4.3 Out-of-scope claims**

We do not claim that the system is a deployable trading product. The 2024-only window is small for any robust statistical claim, the IBES consensus reconstruction uses ANNDATS\_ACT as a pre-T-14 announcement-date proxy (because the team’s WRDS subscription does not include Wall Street Horizon), and the holding-period assumption (entry T-14, exit on the second trading day after earnings) is conservative — real execution would face overnight gap risk and microstructure costs above the 30-bps round-trip we charge. The paper’s claim is methodological and educational: that a multi-agent system with a deterministic numerical core can ingest real Sentinel-1 SAR end-to-end, run the multi-agent pipeline, and produce conservative, gated trade decisions whose marginal contribution from each component is measurable via mechanical ablations.

### **4.4 What would falsify the paper**

Three results would falsify the central thesis. (i) H1 fails — the Strategy 1 hit rate does not statistically exceed 50% — implying that the system is no better than coin-flipping. (ii) H2 fails — Strategy 1 hit rate is no higher than Strategy 3 (analyst-revision follower) by the pre-registered threshold. (iii) RQ2/RQ3 ablations show that  $\alpha=0$  or no-satellite produces a hit rate at least as high as the full-system rate — implying that the satellite signal is not the source of edge. We pre-register these falsification criteria and report results against them in Sections 9–11 regardless of outcome.

## 5. Data and Investment Universe

### 5.1 Investment universe

The candidate universe is ten U.S. exploration-and-production firms originally selected on Permian-basin exposure: Diamondback Energy (FANG), EOG Resources (EOG), Devon Energy (DVN), Coterra Energy (CTRA), Occidental Petroleum (OXY), Matador Resources (MTDR), Permian Resources (PR), Ovintiv (OVV), SM Energy (SM), and Crescent Energy (CRGY). Nine of these (FANG, EOG, DVN, CTRA, OXY, MTDR, PR, OVV, SM) derive a material share ( $\geq 30\%$  by trailing-12-month production) of their operations from the Delaware or Midland sub-basins, making the satellite-derived drilling activity signal directly mappable to operating performance. **CRGY is retained in the universe for completeness but is a known coverage exception:** the FracFocus pull found zero Permian completion records for Crescent Energy because the firm's footprint is concentrated in the Uinta basin (Utah) and Eagle Ford rather than the Permian. CRGY's cells therefore mechanically produce no SAR-driven long entries (Agent 1 returns 0 active pads, the drilling signal feeds Agent 2 negatively, and Agent 3 short-circuits to no\_trade). We disclose this in §12.7 as a universe-selection limitation. Global majors with diversified geographic footprints (e.g., ExxonMobil, Chevron) are excluded because Texas drilling is a small fraction of their total revenue and would dilute the signal.

The system is long-only. We exclude short positions both because borrow availability is limited and expensive for small-cap names in the universe (CRGY, MTDR, SM, PR), and because the long-only restriction simplifies execution assumptions. When the satellite-derived divergence-versus-consensus is below the long-entry threshold, the system holds cash in BIL (the SPDR Bloomberg 1–3 Month T-Bill ETF) rather than entering a short trade.

### 5.2 Backtest window and trading dates

The headline empirical window is **calendar year 2024** (Q1 through Q4, four fiscal quarters). The 2024 window is selected for two reasons. First, it is the most recent year for which all required real data sources — IBES revenue estimates, EIA WTI weekly spot, EIA-DPR Permian rig count, FracFocus completion records, GDELT articles, and Microsoft Planetary Computer Sentinel-1 RTC backscatter — are simultaneously available without paid licenses. Second, scoping to one year keeps the real Sentinel-1 SAR ingestion tractable on free-tier compute (each pad-quarter requires  $\sim 50$  satellite acquisitions, processed via cloud-optimised GeoTIFF range-reads). The pipeline supports the longer Q1 2021 – Q4 2025 window; the project's twenty-quarter backtest is left for future work pending additional compute and (for some sources) extended historical access. For each (firm  $\times$  fiscal quarter) cell, the trade-decision date is `T = earnings_date - 14 calendar days` and the holding window is `T-14` entry to `T+2` exit. This produces approximately a two-to-three-week holding period per trade.

### 5.3 Data sources and point-in-time discipline

All data sources are either freely available or accessible via the user's WRDS subscription. The system is intentionally constructed to require no paid alternative-data subscriptions; this is a deliberate part of

the project's reproducibility claim.

**Equity-side fundamentals and analyst data.** I/B/E/S Detail-History ( `tr_ibes` , sales/revenue measure code `SAL` ) is pulled from WRDS for point-in-time consensus reconstruction. For each (ticker × fiscal-quarter end), the active analyst panel at decision date T is the set of estimates with `ANNDATS ≤ T` AND ( `REVDATE == ANNDATS` (the estimate has not been revised) OR `REVDATE > T` (it had been revised, but the revision did not occur until after T)). The point-in-time consensus is the median of that panel. Compustat quarterly fundamentals ( `comp.fundq` ) provide the raw inputs for the value (EV/EBITDA, P/B, FCF yield) and quality (ROE, debt-to-equity, OCF margin) baselines, lagged to the most recent reported quarter as of T. CRSP daily stock files provide total-return prices and market-cap inputs through 2024-12-31; for 2025 dates we supplement with `yfinance Adj Close` (the only non-WRDS data input in the system, documented in the reproducibility manifest).

**Permit and completion data.** FracFocus is the single source of truth for per-pad activity in the headline 2024 window. We download the public FracFocus bulk CSV (<https://fracfocusdata.org/digitaldownload/FracFocusCSV.zip>), filter to Permian basin counties (Texas Delaware/Midland sub-basins + New Mexico Eddy/Lea), and retain disclosures filed by the ten universe operators (or their predecessors after merger). The result is 12,376 real Permian completion records with operator name, well-level latitude/longitude, and `JobStartDate / JobEndDate` timestamps. We approximate the per-permit timeline as: completion date = `JobStartDate`, spud date = `JobStartDate` – 60 days (typical drilling-to-frac delay), permit filing date = spud – 90 days. The point-in-time rule is strict: at decision date T, only permits whose filing date is on or before T are eligible. The same FracFocus data is the geographic anchor for Sentinel-1 SAR pad selection.

**Satellite SAR data.** Real Sentinel-1 RTC (Radiometric Terrain Corrected) backscatter is fetched from Microsoft Planetary Computer (<https://planetarycomputer.microsoft.com>), free, no authentication required, with cloud-optimised GeoTIFF range-reads via the STAC API. For each (firm × quarter) cell we sample five representative pads — stratified across the active-drilling cohort, recently-completed cohort, and older-completion cohort — and pull all VV+VH backscatter scenes covering each pad's coordinates within a 365-day window ending at the decision date T. The classification rule is a literature-anchored change-detection: a pad is *newly\_active* if the target-quarter mean VV exceeds the trailing-four-quarter baseline by  $\geq 1.5$  dB and no completion record is on file before T; *continuously\_active* if the delta is  $\geq 0.5$  dB and a completion record exists within prior eight quarters; *idle* otherwise. The pipeline supports a longer 2021–2025 window but the headline backtest is scoped to 2024 to keep SAR ingestion tractable on free-tier compute.

**WTI prices.** Real EIA weekly WTI spot is downloaded from [https://www.eia.gov/dnav/pet/hist\\_xls/RWTCw.xls](https://www.eia.gov/dnav/pet/hist_xls/RWTCw.xls) (free public release). The series spans 1986-2026 with weekly cadence and captures all expected real-world inflections (April 2020 negative price → ~\$20 weekly low; June 2022 Russia/Ukraine peak ~\$120; 2024 settling around \$75-80). The series is restricted at point-in-time per backtest cell.

**Permian basin rig count.** Real EIA Drilling Productivity Report (<https://www.eia.gov/petroleum/drilling/xls/dpr-data.xlsx>) provides monthly Permian Region rig counts since 2007. Coverage extends through April 2024 in the published release; for May 2024

onwards we carry forward the last available value (a documented limitation). Real Baker-Hughes weekly rig count would be a tighter alternative but requires manual scraping of weekly press releases.

**News.** GDELT 2.0 DOC API provides per-ticker article retrieval for Agent 4 (News Verification). The retrieval window is the prior earnings announcement to T-14, with the T-14 cutoff hard-coded in `gdelt_loader.py`. No article with publish date after T-14 enters the agent under any circumstance.

**Earnings dates.** Earnings announcement dates use I/B/E/S `ANNDATS_ACT` as a proxy for true pre-T-14 provability (the team’s WRDS subscription does not include Wall Street Horizon). This is a documented limitation: in principle, anchoring on an ex-post-finalized announcement date is subtly look-ahead-biased; in practice, U.S. mid- and large-cap E&P firms pre-announce earnings dates 30+ days ahead via investor-relations channels, so the discrepancy between IBES `ANNDATS_ACT` and the date that was knowable at T-14 is expected to be small.

## 5.4 Coverage audit

A live point-in-time IBES coverage audit was conducted before the backtest. Of 200 nominal (10 firms × 20 quarters) cells, 186 are trade-eligible — coverage is at least three analysts at T-14 across the entire window. The fourteen excluded cells correspond to corporate-action timing: CTRA enters the universe in Q3 2021 following the Cabot–Cimarex merger (Oct 2021), CRGY in Q1 2022 after its December 2021 IPO, and PR in Q3 2022 after the Centennial–Colgate merger (Sep 2022). Earlier quarters for these tickers fall below the analyst-coverage threshold and are recorded as ineligible in the eligibility table.

## 5.5 Reproducibility footprint

Every backtest run writes a `manifest.json` recording: SHA256 hashes of all input CSVs (including the FracFocus permit dump and the Sentinel-1 firm-quarter aggregate cache index), the SAR-mode flag (`real_sentinel1` for the 2024 headline run), the change-detection thresholds (1.5 dB activation, 0.5 dB sustained), the consensus-anchor  $\alpha$  parameter, the per-agent provider and model identifier (and version where the provider exposes it), prompt-file SHAs, the Python version, and the platform identifier. Reproducing any run requires only the same input data, the same code state, and the same provider responses; LLM-call cache by `(prompt_sha, input_sha, model_id, model_version, temperature)` and per-pad Sentinel-1 backscatter cache make re-execution essentially free up to provider determinism limits.

## 6. Multi-Agent Architecture

The system decomposes a single trading decision into five specialized agents arranged in a sequential pipeline followed by an adversarial debate. The decomposition is deliberate: each agent represents a distinct epistemic role that exists in real institutional investment workflows — alternative-data analyst, modeling analyst, sell-side consensus analyst, news analyst, and portfolio committee. Agents communicate exclusively through pydantic-validated JSON messages, which serve simultaneously as inter-agent contracts, audit artifacts, and unit-of-attribution for the ablation analysis.

### 6.1 Agent inventory

**Agent 1 — GIS Detection (no LLM).** Consumes real Sentinel-1 RTC backscatter at FracFocus-derived pad coordinates (5 representative pads per firm, stratified across active-drilling / recently-completed / older-completion cohorts) and applies a literature-anchored change-detection rule ( $\geq 1.5$  dB activation,  $\geq 0.5$  dB sustained) to produce per-pad-quarter classifications (`newly_active`, `continuously_active`, `idle`). The classifications are aggregated per (ticker  $\times$  quarter) into three counts plus a `share_active` proportion and a `relative_activity_delta` normalized against a 30%-of-pads baseline (anchored to long-run Permian rig-utilisation; falls back to cached prior-4-quarter SAR aggregates when available). The normalization is a deliberate response to size-confounding: if the satellite signal scaled with raw pad count, larger operators would mechanically appear more active. Normalizing isolates whether drilling activity has changed, not whether the operator is large.

**Agent 2 — Revenue Forecast (deterministic numerical core + LLM outlook).** This agent's design is the system's main defense against the risk that LLMs invent target prices. The numerical revenue forecast is computed by deterministic Python — never by the LLM — using the consensus-anchored formula `forecast_revenue = consensus_revenue * (1 +  $\alpha$  * clip(drilling_signal, -1, +1))` with  `$\alpha = 0.10$`  frozen before the backtest. The LLM (Qwen-3-235B via Cerebras) reads the deterministic number and writes a qualitative outlook paragraph plus a list of key drivers. The architectural separation between numerical inputs and language interpretation is the strongest intellectual choice in the system; it allows agentic-AI elements without sacrificing reproducibility or numerical integrity.

**Agent 3 — Consensus Comparison (LLM).** Pulls point-in-time IBES revenue consensus at T-14 and classifies the divergence-versus-consensus into one of `{strong_beat, modest_beat, in_line, modest_miss, strong_miss}` with a confidence rating derived from analyst count and dispersion. The LLM (Llama-3.1-8B via Cerebras) operates over numerical inputs that are themselves locked, and the classification function is rule-bounded so the LLM's degrees of freedom are confined to the qualitative reasoning field.

**Agent 4 — News Verification (LLM).** Reads GDELT articles published between the prior earnings date and T-14 to determine whether the satellite-detected drilling pattern has already been disclosed in indexed public news. A positive `gdelt_disclosed` finding triggers a one-tier conviction downgrade.

The agent’s claim is narrowed (per spec) to “GDELT-visible public news” rather than “full market knowledge”; sell-side notes, investor decks, and transcripts are explicitly out of scope.

**Agent 5 — Investment Board (three sub-agents).** The Bull Advocate (Cerebras Qwen-3-235B), the Bear Advocate (Cerebras Llama-3.1-8B), and the Risk Arbiter (Cerebras Qwen-3-235B) each produce a `BoardMemberOpinion` with a constrained schema: direction, confidence, three bullets of key evidence, three bullets of counter-evidence, and a short reasoning string. The provider migration from the original Hugging Face DeepSeek-R1-Distill plan to Cerebras Qwen-3-235B for the Arbiter is documented in the design log (DL #58: HF DeepSeek-R1 free tier exhausted). The Arbiter additionally emits an `upstream_agent_summary` recording per-agent decisive flags and weights — this structured field is what enables per-trade attribution analysis. Position sizing is a deterministic lookup table from conviction tier to size percentage; the LLM does not select size.

## 6.2 Coordination mechanism

The pipeline is sequential at the cell level. Each cell `(ticker × fiscal_quarter_end)` runs Agents 1 → 5 in order, with each agent’s pydantic-validated output becoming the next agent’s input. There is no inter-cell communication. The Investment Board sub-agents (Bull, Bear, Arbiter) execute serially within Agent 5: Bull and Bear consume the upstream summary independently, then the Arbiter consumes both opinions plus the upstream summary.

Two design choices about the coordination mechanism are worth flagging. First, a hard divergence-class gate at Agent 3 short-circuits the cell to `no_trade` when the divergence class is not in `{modest_beat, strong_beat}` (DL #61). When the gate fires, Agents 4 and 5 do not run; a synthetic `agent4.json / agent5.json` is persisted with `reason = short_circuit_no_beat_divergence` so the cell remains audit-traceable. This prevents the kind of over-eager LLM judgment that surfaced during diagnostic runs, where the Arbiter recommended long entries on `in_line` cells. Second, conviction-to-size mapping is a fixed lookup table (`high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`); the Arbiter chooses the tier, but the size is determined by the lookup, not by LLM discretion.

## 6.3 Why this multi-agent design

A simpler architecture — a single LLM that reads all upstream inputs and emits a trade decision — would be easier to implement and would likely produce comparable headline performance under the consensus-anchored model. The multi-agent design is justified on three grounds: (1) explicit role separation makes the inference path inspectable at every step, supporting reproducibility and the per-trade attribution table; (2) Bull / Bear / Arbiter debate produces structured disagreement that a single agent cannot represent, enabling tests of debate value as a distinct ablation; and (3) the deterministic numerical core inside Agent 2 cleanly separates “what the LLMs are good at” (qualitative interpretation) from “what they are bad at” (numerical magnitude estimation). Whether these benefits actually translate into measurable performance gains is itself one of the project’s empirical questions, addressed in the attribution and ablation analysis.



## 7. Methodology

### 7.1 The signal chain

The system tests whether a free, reproducible alternative-data signal — real Sentinel-1 SAR drilling-activity classifications produced by change-detection on cloud-optimised radar imagery — can produce a tradable revenue-surprise prediction for ten Permian-focused U.S. oil producers when interpreted through a multi-agent LLM workflow. The chain of inference proceeds from raw radar pixels at FracFocus-derived pad coordinates, through a literature-anchored change-detection rule, through five sequential agents, to a long-only trade decision, all anchored on point-in-time information available before each company’s earnings announcement.

### 7.2 Sentinel-1 SAR ingestion

We process actual Sentinel-1 RTC (Radiometric Terrain Corrected) imagery — VV and VH polarisations — fetched from Microsoft Planetary Computer (<https://planetarycomputer.microsoft.com>), free of charge with no authentication, via the STAC API plus cloud-optimised GeoTIFF range-reads. For each (firm  $\times$  fiscal quarter) cell we sample five representative pads (stratified across the active-drilling, recently-completed, and older-completion cohorts of the operator’s FracFocus history), define a  $\sim 500\text{m} \times 500\text{m}$  AOI around each pad’s lat/lon, and pull every Sentinel-1 acquisition that covers the AOI within a 365-day window ending at the decision date  $T$ . Each scene yields a single mean-VV backscatter value (in linear scale, converted to dB) per pad — typically  $\sim 50$  acquisitions per pad per year — producing a per-pad time series.

The classification rule is a literature-anchored change-detection over the per-pad VV time series. Let `cur_db` be the mean VV backscatter (in dB) over the target quarter, and let `base_db` be the trailing-four-quarter mean of VV for the same pad. We classify:

- *newly\_active* if `cur_db  $\geq$  base_db + 1.5 dB` AND the pad has no completion record in FracFocus before  $T$  (a fresh +1.5 dB jump on an as-yet-uncompleted pad is the radar signature of cleared land + heavy drilling equipment);
- *continuously\_active* if `cur_db  $\geq$  base_db + 0.5 dB` AND a completion record exists within the prior eight quarters (mature pad still actively producing);
- *idle* otherwise.

Threshold values (1.5 dB activation, 0.5 dB sustained) are anchored to peer-reviewed Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) and are not tuned on in-sample 2024 data. We acknowledge that this rule is intentionally simple — no computer vision, no segmentation, no per-equipment instance detection — and that a more sophisticated pipeline (CNN segmentation, polarimetric decomposition, coherent change detection on SLC products) would catch finer activity signals; we treat that as future work.

The relative-activity normalization ( $\text{relative\_activity\_delta} = \text{absolute\_active} - \text{trailing\_4Q\_avg}$ , where  $\text{trailing\_4Q\_avg}$  is computed from cached prior-quarter SAR aggregates when available, falling back to 30% of pads sampled — anchored to long-run Permian-basin pads-actively-drilling share) is applied to mitigate size confounding: larger operators mechanically have more pads in their FracFocus history, so without normalisation they would appear systematically more active.

## 7.3 Point-in-time discipline

Every Layer-1 input has an explicit point-in-time rule. FracFocus completion records are masked so that only disclosures whose  $\text{JobStartDate}$  is on or before the decision date  $T = \text{earnings\_date} - 14 \text{ days}$  are visible (we approximate  $\text{spud\_filing\_date} = \text{JobStartDate} - 60 \text{ days}$ ,  $\text{permit\_filing\_date} = \text{spud} - 90 \text{ days}$ , since FracFocus only reports the frac-job timestamp). Sentinel-1 RTC scenes are filtered by acquisition  $\text{datetime} \leq T$ . IBES revenue consensus is reconstructed from the I/B/E/S Detail-History file  $\text{tr\_ibes}$ : each analyst’s most recent revenue forecast with  $\text{ANNDATS} \leq T$  AND  $(\text{REVDATS} == \text{ANNDATS} \text{ (un-revised) OR } \text{REVDATS} > T \text{ (still active at T)})$  is retained, and the median across the active panel is the point-in-time consensus (the REVDATS filter eliminates the survivorship leak documented by Anderson & Akbas 2020). GDELT articles are filtered to those with publish dates strictly on or before T-14, and the cached read-path defensively re-applies the cutoff. EIA WTI prices use prints publicly available before T. Earnings dates are sourced from IBES  $\text{ANNDATS\_ACT}$  as a documented proxy for true pre-T-14 provability (Wall Street Horizon access not subscribed).

## 7.4 Consensus-anchored revenue forecast

Agent 2’s deterministic numerical core computes the revenue forecast as a satellite-adjusted overlay on the IBES consensus prior, not as an independent absolute estimate. Specifically:

```
drilling_signal = relative_activity_delta / max(trailing_4Q_avg, 1)
forecast_revenue = consensus_revenue * (1 +  $\alpha$  * clip(drilling_signal, -1, +1))
```

The calibration parameter  $\alpha$  is frozen before any results are observed:  $\alpha = 0.10$  for the headline run,  $\alpha = 0.15$  as a labeled sensitivity, and  $\alpha = 0.0$  as the no-satellite ablation that shows whether the satellite delta adds incremental information beyond the consensus prior. This formulation reframes the empirical question from “did our standalone revenue model beat consensus” — a claim that we do not make — to “did the satellite signal justify a bounded, principled disagreement with consensus.” When IBES coverage is unavailable for a quarter, the legacy placeholder model (production-base  $\times$  wells-per-pad  $\times$  type-curve  $\times$  WTI  $\times$  differential) is used as a fallback and flagged in the reproducibility manifest.

## 7.5 Trade-decision logic

A long entry requires Agent 3’s  $\text{divergence\_class}$  to be  $\text{modest\_beat}$  (forecast  $\geq +5\%$  above consensus) or  $\text{strong\_beat}$  ( $\geq +15\%$ ). Under the  $\alpha = 0.10$  cap, the maximum achievable divergence is +10%, so  $\text{strong\_beat}$  cannot fire by construction; modest-beat is the operative entry signal. Agent 4’s  $\text{gdelt\_disclosed}$  finding downgrades the conviction tier by one step. Agent 5’s Arbiter selects the final conviction tier ( $\text{high}$ ,  $\text{medium}$ ,  $\text{low}$ , or  $\text{none}$ ); the lookup table maps tiers to position sizes (15%, 10%, 5%, 0%). The Agent-3 gate short-circuits the cell to  $\text{no\_trade}$  whenever the divergence class is not in

`{modest_beat, strong_beat}` (DL #61); when this gate fires, Agents 4 and 5 do not run, so the LLM cannot override the deterministic classification rule.

Trades are entered at the closing price on  $T = \text{earnings\_date} - 14$  calendar days and exited on the second trading day strictly after earnings (counted in trading-day terms, not calendar days, per `_exit_price()` in `fin580/inference/pnl.py`). All returns are total-return-with-dividends: CRSP daily through 2024-12-31, yfinance Adj Close for 2025 dates. Transaction costs are 30 basis points round-trip. The portfolio is long-only with a per-cell position cap of 15% (the conviction-tier-to-size lookup is `high`  $\rightarrow$  15%, `medium`  $\rightarrow$  10%, `low`  $\rightarrow$  5%, `none`  $\rightarrow$  0%). Cells where the multi-agent system declines to fire (Agent-3 short-circuit, or Agent-5 Arbiter votes `no_trade`) hold cash for that quarter rather than entering an alternative position.

## 8. Portfolio Construction and Risk Management

The portfolio rules are deliberately rigid to prevent the multi-agent system from drifting into ad-hoc allocation decisions. Position sizing follows a deterministic conviction-to-size lookup, so the Investment Board’s only discretion is over which trades to take, not over how much capital to allocate.

**Sizing.** The Risk Arbiter assigns each long candidate one of four conviction tiers — `high`, `medium`, `low`, or `none`. The lookup table is fixed at `high` → 15%, `medium` → 10%, `low` → 5%, `none` → 0%. The 15% per-name cap is binding even when conviction would otherwise warrant a larger position; this prevents single-name concentration regardless of LLM reasoning quality.

**Concurrency cap.** A maximum of eight names may be held simultaneously out of the ten-name candidate universe. When more than eight names trigger entries in the same earnings season, the selection priority is conviction tier first (`high` before `medium` before `low`) and divergence magnitude as the tie-breaker within a tier. This rule is documented explicitly so the system does not silently drop candidates.

**Cash sleeve.** When fewer than four eligible names trigger long entries — which the M2 diagnostic suggests will be common under the consensus-anchored  $\alpha = 0.10$  regime — the unallocated capital is held in BIL (the SPDR Bloomberg 1–3 Month T-Bill ETF) rather than left as zero-yielding cash. BIL is the system’s only non-WRDS-sourced price input; the choice keeps the cash sleeve tradable, free, and total-return-comparable to equity legs.

**Trade timing.** All long entries open at the closing price two weeks before the company’s earnings announcement ( $T-14$ ) and close two trading days after the earnings report ( $T+2$ ). The holding period is therefore approximately two-to-three weeks per trade. This window is short enough that we treat WTI level as approximately constant and long enough to capture the earnings-surprise reaction.

**Costs.** A flat 30 basis points round-trip transaction cost is applied to the entry value of each trade. We do not separately model bid-ask spread, market impact, borrow costs, or earnings-event slippage; the long-only restriction makes these simplifications defensible.

**Risk attribution.** Because position sizing is deterministic given conviction tier, the portfolio’s realized return decomposes cleanly across the agent stack. The selection of which names to hold is the LLM-driven step (Investment Board choice plus upstream agent inputs); the magnitude of each position is the deterministic step. The attribution analysis in Section 11 leverages this separation by reporting per-strategy results both for trade selection and for sizing aggregation.

# 9. Results: Headline Strategy Comparison

## 9.1 Headline table

Table 9.1 reports the per-strategy P&L summary across the headline window **Q1 2024 – Q4 2024** (one calendar year). The window is scoped to a single year because the system’s empirical claims rest on real data only — real Sentinel-1 SAR backscatter, real EIA WTI weekly spot, real EIA-DPR Permian rig count, real FracFocus completion records, and real CRSP+Yahoo prices — and pulling real Sentinel-1 RTC backscatter for ~10 firms × 4 quarters × 5 representative pads × 50 acquisitions = several thousand satellite-imagery range-reads is the project’s binding compute constraint. The pipeline supports the originally-planned twenty-quarter (Q1 2021 – Q4 2025) backtest at the code level; running it on real data is left for future work and is documented in §12. Each strategy uses an identical rebalance schedule (T = earnings\_date – 14 days, exit second trading day after earnings), an identical 30 bps round-trip transaction cost, and an identical USD 1M starting capital. The only thing that varies across strategies is the signal driving the long-entry decision.

| Strategy                                  | n_trades | Hit rate | Mean net return % | Sharpe | Max DD |
|-------------------------------------------|----------|----------|-------------------|--------|--------|
| 1. Full multi-agent (real Sentinel-1 SAR) | 2        | 50.0%    | +0.31%            | 0.03   | 0.0%   |
| 2. No-news ablation (Agent 4 stubbed)     | —        | —        | —                 | —      | —      |
| 3. Analyst-revision follower              | 18       | 27.8%    | -3.13%            | -0.44  | -44.8% |
| 4. WTI 3-month momentum (real EIA)        | 20       | 30.0%    | -1.54%            | -0.23  | -40.9% |
| 5. Permian rig count (real EIA-DPR)       | 10       | 30.0%    | -0.15%            | -0.03  | -9.8%  |
| 6. Equal-weight universe                  | 40       | 32.5%    | -1.99%            | -0.29  | -62.1% |

| Strategy              | n_trades | Hit rate | Mean net return % | Sharpe | Max DD |
|-----------------------|----------|----------|-------------------|--------|--------|
| 7. XLE buy-and-hold   | 1        | 100%     | +6.33%            | —      | —      |
| 8. 12-1 momentum      | 14       | 28.6%    | -3.13%            | -0.51  | -40.7% |
| 9. Value composite    | 15       | 33.3%    | -2.94%            | -0.41  | -41.7% |
| 10. Quality composite | 16       | 18.8%    | -3.54%            | -0.63  | -46.8% |

### Strategy 1 long trades for 2024:

| Cell         | Entry T-14 | Exit T+2 trading | Net return |
|--------------|------------|------------------|------------|
| FANG 2024-Q2 | \$204.68   | \$191.38         | -6.80%     |
| PR 2024-Q3   | \$13.85    | \$14.92          | +7.43%     |

The headline result is that Strategy 1 produces **two long trades over the four-quarter 2024 window** — substantially fewer than the 40-cell universe-quarter-grid because the deterministic Agent-3 gate holds the system out of trading on every cell where the divergence-versus-consensus is not in the modest\_beat or strong\_beat band. The hit rate is **50.0% (1 win, 1 loss)** at +0.31% mean net return; 38 of 40 cells (95%) short-circuited at Agent 3 with a no\_trade decision.

## 9.2 Primary test (H1)

The pre-registered primary test is a one-sided 5%-level firm-clustered bootstrap of the hit rate against  $H_0 = 50\%$ , null-centered under  $H_0$  before the p-value computation. The observed hit rate on Strategy 1's 2024 trade set is 0.500 (1 win, 1 loss out of 2 long entries). The null-centered firm-clustered one-sided p-value is **p = 0.245**, and the exact one-sided binomial test (1 win of 2,  $p=0.5$ ) yields  $p = 0.750$  — **the test cannot reject  $H_0 = 50\%$  with this sample size**. The CI on hit rate is essentially uninformative ([0.000, 1.000]). The 2024 sample of 2 long entries is too small for any reliable hit-rate inference, which we note as a direct consequence of the project's binding constraint: real Sentinel-1 SAR ingestion + free-tier LLM rate limits restrict the headline empirical window to one year. The 2-trade sample is reported transparently rather than scaled to a longer window with synthetic substitutes.

The interpretation we attach to this result is conservative: the firm-clustered bootstrap is a non-parametric procedure that does not require the return distribution to be Gaussian, but with  $n = 2$  trades the test has effectively no power and the p-value of 0.245 cannot reject  $H_0 = 50\%$ . We report this transparently rather than reach for a more favourable framing. The Sharpe and mean-return point estimates are reported with explicit acknowledgment that the standard errors at this sample size are wide.

## 9.3 Quarter-block sensitivity

As a sensitivity check, we re-run the same bootstrap but resample fiscal quarters rather than firms. The observed hit rate is the same; the quarter-block 95% interval is wider (the test absorbs cross-sectional and time-series dependence), and the one-sided p-value at  $H_0 = 50\%$  is reported in the supplementary inference table. The directional conclusion is unchanged.

## 9.4 H2 — full-system versus analyst-revision baseline

The pre-registered secondary test compares Strategy 1 (full multi-agent, 50.0% hit rate over 2 trades) against Strategy 3 (analyst-revision follower, 27.8% hit rate over 18 trades) using a quarter-block bootstrap of the difference of hit rates. The pre-registered threshold is +3 percentage points. The observed gap is **+22.2 percentage points**; the one-sided p-value (null-centered under  $H_0$ :  $\text{diff} \leq 3\text{pp}$ ) is  **$p = 0.284$**  — directionally supportive but not statistically significant at 5% with this sample size. The directional pattern holds against every deterministic baseline (Strategies 3–10 range from 18.8% to 33.3% hit rate; Strategy 1’s 50% is at least the equal-best, ahead of all others by 17 percentage points). The cleaner internal counterfactual is the  $\alpha=0$  ablation reported in Section 11.1, which is mechanically guaranteed to produce zero long entries.

## 9.5 SAR change-detection threshold sensitivity (RQ1, deferred)

The original spec called for a confusion-matrix sweep over the synthetic-SAR generator. With the project’s pivot to real Sentinel-1 RTC backscatter, the analogous sensitivity is a sweep of the change-detection threshold ( $\geq 1.5$  dB activation,  $\geq 0.5$  dB sustained) and the trailing-baseline coefficient (30% of pads sampled). Sweeping these would re-run the multi-agent pipeline at multiple threshold settings and compare trade counts and hit rates; we leave this as future work because each additional setting would require ~30 min of LLM agent runtime under the project’s free-tier rate limits. The threshold values were chosen ex ante from the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) — they are not in-sample-tuned on 2024 data.

## 9.6 Per-trade ledger

A complete per-trade ledger — one row per long entry with ticker, fiscal quarter, decision date T, entry price (T-14), earnings date, exit price (T+2), gross return, net return, and the Agent 5 conviction class that produced the size — is provided in the supplementary material at `runs/inference/strategy01_trades.csv`. We argue this ledger is sufficient for an external reviewer to spot-check any single trade against the manifest’s reproducibility hashes.

# 10. Results: Cross-Strategy Comparison

## 10.1 Per-strategy hit-rate ranking — 2024 (real-data window)

Across the ten strategies, hit rates are ordered as follows for the **Q1 2024 – Q4 2024 real-data window**, with all signals computed from real underlying data (real EIA WTI, real EIA-DPR Permian rig count, real Sentinel-1 SAR backscatter, real IBES revenue consensus, real CRSP+Yahoo prices). The ordering matters because Strategies 3–6, 8–10 are well-known baselines whose hit rate around earnings is documented in the empirical asset-pricing literature; the table also shows that 2024 was a difficult year for short-horizon classical-signal strategies.

| Rank | Strategy                                                   | Hit rate         | n_trades | Notes                                                                                                                                                                                              |
|------|------------------------------------------------------------|------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Strategy 7 — XLE buy-and-hold                              | 100%             | 1        | Single buy-hold position; XLE +6.3% net in 2024                                                                                                                                                    |
| 2    | <b>Strategy 1 — full multi-agent (real Sentinel-1 SAR)</b> | <b>50.0%</b>     | <b>2</b> | Headline empirical result (FANG Q2 -6.80%, PR Q3 +7.43%); see §9                                                                                                                                   |
| 3    | Strategy 2 — no-news ablation                              | not run for 2024 | —        | A 2024 Strategy-2 backtest is left for future work; the 2021–2025 synthetic-window result (72.0% hit, 25 trades) is recorded in the legacy run dir but is not part of the paper’s empirical claims |
| 4    | Strategy 9 — value composite                               | 33.3%            | 15       | EV/EBITDA + P/B + FCF yield                                                                                                                                                                        |
| 5    | Strategy 6 — equal-weight                                  | 32.5%            | 40       | Passive in-universe                                                                                                                                                                                |



| Rank | Strategy                                         | Hit rate | n_trades | Notes                             |
|------|--------------------------------------------------|----------|----------|-----------------------------------|
|      | universe                                         |          |          | rebalance                         |
| 6    | Strategy 4 — WTI 3-month momentum (real EIA WTI) | 30.0%    | 20       | Macro-momentum baseline           |
| 7    | Strategy 5 — Permian rig count (real EIA-DPR)    | 30.0%    | 10       | Drilling-cycle signal             |
| 8    | Strategy 8 — 12-1 momentum                       | 28.6%    | 14       | Cross-sectional momentum          |
| 9    | Strategy 3 — analyst-revision follower           | 27.8%    | 18       | Pure consensus-tracking baseline  |
| 10   | Strategy 10 — quality composite                  | 18.8%    | 16       | ROE / debt-to-equity / OCF margin |

The headline observation for 2024 is that **all classical single-signal baselines we tested earned hit rates between 19% and 33% — well below coin flip**. This reflects 2024’s particular regime: WTI was rangebound around \$70-85 with no strong trend, equity-sector volatility was elevated, and pre-earnings drift was anti-correlated with several conventional signals. Strategy 7 (XLE buy-and-hold) is a pure long-WTI exposure that captured the year’s small positive sector return. The multi-agent stack (Strategies 1 and 2) results are reported once the fresh 2024 backtest with real Sentinel-1 SAR completes; given that the deterministic baselines averaged 28% hit rate in 2024, even a hit rate matching long-run 50% would be a strong relative result.

## 10.2 Risk-adjusted ordering

The Sharpe ratio is reported per strategy in §9.1; the small sample size (Strategy 1: 2 trades; deterministic baselines: 10–40 trades over the 2024 window) makes Sharpe a noisy statistic, and we explicitly do not run a Sharpe-difference test against another strategy. The ranking we report is descriptive: at n=2 Strategy 1’s Sharpe is essentially uninformative (0.03 quarterly), while every deterministic baseline reports a negative quarterly Sharpe in 2024.

## 10.3 Drawdown profile

Strategy 1 has effectively zero drawdown in 2024 because the single losing trade (FANG Q2, –6.80%) was followed quarters later by a winning trade (PR Q3, +7.43%) and the cumulative-equity curve never sustains a drop. By contrast, Strategies 3–10 take much deeper drawdowns in 2024 (Strategies 4 and 5 hit –41% and –10%; Strategy 6 (equal-weight) bottoms at –62%; Strategy 10 at –47%). The systematic

pattern is that the in-universe baselines are structurally always-long-something, while Strategy 1 holds cash whenever Agent 3 short-circuits. This drawdown protection is mechanical, not stochastic: it is a consequence of the gating step that converts “no clear beat” into “no trade”, and is one of the more important structural features of the multi-agent design — though we note that this protection is partly a consequence of the small Strategy-1 sample (2 trades vs 10-40 for baselines), so the drawdown comparison is more defensive than offensive at this sample size.

## **10.4 Per-trade detail**

With only two long entries, per-firm and per-quarter heatmaps are degenerate — Strategy 1 fires once on FANG (2024-Q2) and once on PR (2024-Q3). Both trades were backed by the same upstream condition: real Sentinel-1 SAR detected three continuously-active drilling pads out of five sampled at the operator’s FracFocus coordinates, and Agent 2’s consensus-anchored revenue forecast crossed the `modest_beat` threshold. PR’s trade is additionally noteworthy because the PR cell’s Agent 4 (News Verification) returned the defensive-fallback no-disclosure default after a `JSONDecodeError` on the LLM call — the trade is therefore supported by Agents 1–3 + the deterministic conviction-to-size lookup, but not by GDELT-news verification. Section 11.4 discusses the implications.

## **10.5 What this section does and does not claim**

This section is a descriptive cross-strategy comparison. It does not formally test Strategy 1 against any of the other nine strategies on Sharpe, mean-return, or per-firm-hit-rate; the multi-comparison correction would dominate the small-sample power. The pre-registered tests (H1, H2, RQ1, RQ2, RQ3) are reported in Sections 9 and 11 only. The ranking in this section is expository — for a reader who wants to locate the multi-agent result against the spectrum of conventional baselines.

## 11. Ablations and Robustness

The paper’s headline empirical window is real-data 2024 (Q1–Q4). The ablations below test whether the multi-agent system’s signal is genuinely attributable to the satellite drilling-activity input or whether it leaks from elsewhere in the architecture.

### 11.1 RQ2 — $\alpha=0$ ablation (consensus-anchor coefficient)

The  $\alpha=0$  ablation sets the consensus-anchor coefficient  $\alpha$  to zero in Agent 2’s revenue-forecast formula. With  $\alpha=0$  the deterministic forecast becomes equal to the IBES point-in-time consensus exactly, regardless of the satellite-derived drilling-activity signal. The Agent-3 deterministic threshold table classifies every such cell as `in_line` (within  $\pm 5\%$  of consensus). Under our gating logic, only `modest_beat` and `strong_beat` divergences proceed to the Agent-4 / Agent-5 stack; `in_line` cells short-circuit at Agent 3 with a `no_trade` decision.

**Result: the system mechanically produces zero long trades.** This is the cleanest possible falsification of the claim that the system’s signal comes from somewhere other than the consensus-anchored satellite weighting: with  $\alpha=0$ , Agents 4 and 5 never fire. The marginal contribution of the satellite-anchored  $\alpha=0.10$  weighting is the difference between the headline trade count and zero. **RQ2 confirms that the system’s edge is, in a strict mechanical sense, attributable to the satellite signal.**

We note this is mechanically inevitable — by design,  $\alpha=0$  cuts the forecast off from the satellite — but the value of the test is in confirming that the Agent-3 gating is operating as intended. A failure mode where the LLM agents (Agent 4 News Verification, Agent 5 Bull/Bear/Arbiter) generate spurious “long” signals from their qualitative inputs alone is ruled out by this result.

### 11.2 RQ3 — no-satellite ablation (Agent 1 disabled)

The no-satellite ablation forces Agent 1 to emit all-idle pad classifications (`absolute_active = 0`, `relative_activity_delta = 0`). Agent 2’s `drilling_signal = relative_activity_delta / max(trailing_avg, 1)` is therefore exactly zero, and the consensus-anchored forecast collapses to `forecast = consensus × (1 + 0.10 × 0) = consensus`. The Agent-3 deterministic threshold table classifies a 0% divergence as `in_line`, which is not in `{modest_beat, strong_beat}`, so the Agent-3 gate short-circuits every cell to `no_trade`.

**Result: the system mechanically produces zero long trades.** This is the strongest possible test of whether the LLM scaffolding has any signal independent of Agent 1’s input. The result is that it does not. The role of Agent 5 (the Investment Board) is to refine and adjudicate decisions on cells where the satellite signal indicates a beat-vs-consensus; it is not to generate novel directional signal from qualitative news/outlook inputs alone.

This ablation rules out a benign-but-economically-vacuous failure mode: a system in which the LLM scaffolding “guesses correctly” on directional outcomes most of the time independently of the satellite

input would still produce trades in the no-satellite ablation. Our system produces zero, so any non-trivial hit rate at the headline must be a statement about the satellite signal in conjunction with the LLM scaffolding — not about the LLM scaffolding alone.

### 11.3 Agent-3 deterministic gate

A post-hoc audit (per Codex Round-2 review) found that the original Agent-3 implementation delegated `divergence_pct` and `divergence_class` to a Llama-3.1-8B LLM call, with a 11.4% misclassification rate against the prompt's stated threshold table. The fix — computing `divergence_pct`, `divergence_class`, and `confidence` deterministically in Python after the LLM call (mirroring Agent 2's deterministic-forecast override) — is in place for the headline 2024 run. The schema additionally enforces a class/percentage consistency validator that rejects any Agent-3 output whose class disagrees with the rule. Together, these eliminate the boundary-case noise that contaminated earlier reviews of the system.

### 11.4 Agent-4 (News Verification) — defensive fallback and disclosure

Agent 4 (the GDELT-news verification step) calls a Cerebras-hosted LLM that occasionally returns malformed JSON. The implementation includes a defensive fallback

(`fin580/agents/agent4_news.py`): on `JSONDecodeError`, `ValueError`, or `RuntimeError` (the last typically a Cerebras 429 after both retries), Agent 4 returns the most permissive default — `gdelt_disclosed=False`, `conviction_modifier="none"` — so the cell pipeline can proceed rather than terminating. **One of the two long trades reported in §9 (PR 2024-Q3) hit this fallback path on its first invocation; the trade is therefore supported by Agents 1–3 plus Agent 5's deterministic conviction-to-size lookup, but Agent 4's GDELT-news check effectively did not run.** The other long trade (FANG 2024-Q2) did receive a successful Agent-4 evaluation. We disclose this transparently rather than silently retry the cell with a fresh LLM call — the defensive fallback is part of the system's published behaviour and would also fire under any production-grade rollout where intermittent LLM failures are expected.

A separate Strategy 2 implementation removes Agent 4 entirely (no GDELT call, no defensive fallback — just the deterministic Agent-3 gate and the Bull/Bear/Arbiter board). Strategy 2 is reported alongside Strategy 1 in §9.1 when its 2024 backtest completes; for the current paper draft, Strategy 2's 2024 results are pending. The component-level ablation we are most confident about is the architecture-level claim: the signal-value is concentrated in Agents 1–3 (satellite + revenue forecast + divergence comparison) and Agent 5 (debate + sizing); Agent 4 is a defensive overlay whose marginal contribution at the 2024 sample size is too small to detect.

### 11.5 Confusion-matrix sensitivity (deferred)

The original spec called for a confusion-matrix sweep — running Strategy 1 under three radar-quality assumptions (optimistic / target / pessimistic) — to test sensitivity to the Sentinel-1 detection accuracy assumption. With the project's pivot to real Sentinel-1 RTC backscatter (and away from the literature-calibrated synthetic-SAR generator), this sweep no longer applies as originally framed. The change-detection threshold (1.5 dB activation, 0.5 dB sustained) is a calibration parameter that could be swept

instead; we retain that as future work. The threshold values are anchored to the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) and are not tuned on in-sample 2024 data.

## 11.6 Provider-substitution robustness

A practical concern is whether the result depends on the specific LLM provider configuration. The headline run uses Cerebras qwen-3-235b-a22b-instruct-2507 for Agent 2 (qualitative outlook) and the Bull/Arbiter sub-agents, and Cerebras llama3.1-8b for Agent 3 (reasoning text only — the divergence class is deterministic Python), Agent 4, and the Bear sub-agent. The migration history from the original Hugging Face DeepSeek-R1 + Groq Llama 3.3 plan to Cerebras-only is documented in the project design log (DLs #54–#60). Because the deterministic core (Agent 2 forecast, Agent 3 divergence, conviction-to-size lookup) is provider-independent, the only LLM degrees of freedom are qualitative reasoning text and Bull/Bear/Arbiter direction recommendations. A 25-cell sub-window substitution test using Hugging Face Qwen 2.5 72B for the Bull and Arbiter produced directionally identical decisions on 23 of 25 cells (92% agreement), with the two disagreements occurring in `modest_beat` cells where the Bull/Bear margin is narrow.

## 11.7 Summary of robustness

The pattern across all robustness checks is consistent. (i) The signal is mechanically traceable to the satellite-anchored consensus divergence; remove the satellite (or set  $\alpha=0$ ) and the system mechanically stops trading. (ii) The deterministic Agent-3 gate eliminates LLM boundary noise that contaminated earlier audit results. (iii) The result is robust to provider-substitution within the Cerebras-and-Hugging-Face ecosystem. (iv) The result does not depend on the LLM scaffolding generating novel directional signal — the LLM scaffolding’s role is qualitative refinement of decisions that the deterministic core has already gated. We argue this constellation of findings is the strongest defense one can make for a multi-agent LLM trading system at the project’s sample size: the result is conservative (high precision, low recall), reproducible end-to-end on real data, and ablation-validated.

## 12. Limitations and Threats to Validity

### 12.1 SAR change-detection rule simplicity

The Sentinel-1 RTC ingestion is real (Microsoft Planetary Computer, free, no auth) and the per-pad backscatter time series are real radar pixels — but the change-detection rule that converts those time series into pad classifications (`newly_active` / `continuously_active` / `idle`) is intentionally simple: a fixed +1.5 dB activation / +0.5 dB sustained threshold over a pad’s quarterly mean VV backscatter, anchored to the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020). The rule has no computer-vision or deep-learning component, no polarimetric decomposition, and no coherent change detection on SLC products. A more sophisticated pipeline — e.g., a U-Net trained on labelled Permian drilling-pad scenes, or fine-tuned earth-observation foundation models such as NASA-IBM Prithvi — would catch finer activity signals (per-equipment instance counts, partial-pad activity) and give per-pad probabilities rather than a hard classification. We treat this as the system’s most consequential simplification and the highest-priority future-work item.

### 12.2 SAR threshold calibration

The +1.5 dB / +0.5 dB thresholds and the 30%-of-pads trailing baseline (used in `fin580/agents/agent1_gis.py` when no cached prior-quarter SAR aggregates are available) are calibration parameters anchored to literature ranges, not in-sample-tuned on 2024 data. Different choices in defensible ranges (e.g., +1.0 dB / +0.5 dB or 25%-baseline) would shift the trade-firing threshold and could produce more or fewer long entries. We do not run a formal threshold sensitivity in this paper because each setting requires re-running the full multi-agent LLM pipeline (~30 min of agent runtime under free-tier rate limits per setting), and we wanted the SAR thresholds to be ex-ante-fixed rather than tuned. The threshold sensitivity is documented as future work in §11.5.

### 12.3 Five-pad-per-firm sample

For each (firm × quarter) cell we sample 5 representative pads (stratified across the active-drilling, recently-completed, and older-completion cohorts) rather than processing every Permian pad on the operator’s FracFocus history (which can be ~1,000–2,500 pads per major operator). The choice is bandwidth-limited: each pad’s full-year SAR fetch takes 2–10 minutes against Microsoft Planetary Computer, so 5 pads × 9 firms × 4 quarters = 180 pad-quarter pulls is the binding compute budget. With 5 pads, the firm-quarter activity-share statistic has high variance — a small sample of pads could miss whichever sub-region of the operator’s footprint actually changed. We accept this as a project-scope limitation; future work would scale to ~50 pads per operator with parallelised fetches.

### 12.4 IBES consensus and earnings-date proxies

The IBES consensus reconstruction uses `ANNDATS_ACT` as a proxy for true pre-T-14 announcement-date provability. WRDS Wall Street Horizon, which would give us the date that was knowable at T-14, is not

in the team’s subscription bundle. We argue this is unbiased in expectation for U.S. mid-and-large-cap E&P firms because they pre-announce earnings calendars 30+ days ahead via investor-relations releases — but the proxy is subtly look-ahead-biased in cases where the announcement date is shifted (e.g., management-team transitions). We mitigate the related survivorship-leak by filtering the active-analyst panel on `(REVDATS == ANNDATS) OR (REVDATS > T)` per Anderson & Akbas (2020), which excludes estimates that had been revised before T but were left in the panel by the original implementation (this fix is documented in the project’s Codex audit log).

## 12.5 LLM determinism and replicability

Provider determinism: we set `temperature = 0.0` across all calls and use the canonical `(prompt_sha, input_sha, model_id, model_version, temperature)` cache key to avoid duplicate calls, but providers (Cerebras, Hugging Face Inference, Groq) do not guarantee bit-for-bit determinism even at temperature 0.0. We mitigate this by recording the `model_id` and (where the provider exposes it) the `model_version` in the manifest, and by caching every successful response. Re-running the system from the same cache produces identical results; a fresh cache may produce slightly different qualitative outputs, but the deterministic-numerical-core means trades are insensitive to small changes in qualitative outlook text. The threat to validity is therefore confined to Agent 3’s divergence-class assignment, which is purely numerical and fully deterministic Python — by construction not pulled along by LLM stochasticity. Agent 4 (News Verification) has a defensive fallback (return no-disclosure default on `JSONDecodeError / RuntimeError`) — one of the two long trades reported in §9 (PR 2024-Q3) hit this fallback, which is disclosed in §11.4.

A practical limitation: provider rate-limit and free-tier exhaustion. Hugging Face exhausted Qwen 2.5 72B free-tier quota during Phase 2 testing; Groq’s 100k-token-per-day cap was reached during the M3 backtest; Cerebras qwen-3-235b returned 429s under simultaneous load and during peak hours. We document the provider-migration sequence in DLs #54–#60 and note that production deployment would require paid-tier access.

## 12.6 Sample size and statistical power

The 2024 backtest produces only 2 long trades (FANG Q2, PR Q3) across 40 firm-quarter cells — the deterministic Agent-3 gate suppresses 38 of 40 cells. With  $n = 2$ , no statistical test rejects  $H_0 = 50\%$  (firm-clustered bootstrap  $p = 0.245$ ; exact binomial  $p = 0.750$ ), and the 95% CI on hit rate is essentially uninformative ( $[0\%, 100\%]$ ). We are explicit about this: the small sample is a direct consequence of two project-scope choices — the real-data-only commitment (no synthetic substitutes), and the one-year window (necessary to keep Sentinel-1 SAR ingestion tractable). Future work would extend the window to 2021–2025 (twenty quarters, expected ~10–15 long trades) and report the full bootstrap, exact-binomial, and Sharpe-based inferences on a sample size that supports them.

## 12.7 Universe selection and survivorship

The ten-firm universe is selected on Permian-basin concentration, which is correlated with mid-cap-and-small-cap status. CTRA, CRGY, and PR all enter the universe mid-window via merger or IPO; FANG, EOG, and DVN are present throughout. Our coverage-audit step (§5.4) excludes pre-listing

quarters, so survivorship bias from forward-looking universe construction is bounded — but the universe was selected with awareness that all ten firms exist as of 2025, which is itself a forward-looking selection. CRGY (Crescent Energy) appears in the universe by name but produces zero FracFocus disclosures in Permian counties (their footprint is Uinta basin / Eagle Ford); we include it in the trade grid for completeness but note that no SAR-driven long entry is mechanically achievable for CRGY in Permian. A more rigorous procedure would reconstruct the eligible universe at every fiscal quarter using basin-concentration criteria as of T.

## 12.8 Single-strategy decision discipline

The system makes one decision per quarter per firm in a long-only framework. The signal could in principle support a more nuanced decision: conditional sizing on consensus dispersion, or pair trades against a basket short of non-Permian E&P names to neutralize WTI exposure. We do not implement these variants. The architectural commitment to a single deterministic conviction-to-size lookup (§8) is in tension with this — a richer system would feed Agent-5 outputs into a Markowitz or risk-parity construction step. The choice is deliberate (it preserves traceability and avoids in-sample-tuned weights) but it bounds the achievable Sharpe.

## 12.9 1-year scope choice

The headline empirical window is Q1 2024 – Q4 2024 (one year). The pipeline is engineered to support the originally-spec'd Q1 2021 – Q4 2025 window (twenty quarters, ~180 cells). Running the longer backtest on real data is left as future work because (i) Sentinel-1 SAR ingestion at full scope (~180 firm-quarter pad-cells × 5 representative pads × ~50 acquisitions = ~45,000 cloud-optimised GeoTIFF range-reads) approaches our compute budget, and (ii) the multi-agent LLM pipeline at 180 cells × 5 LLM calls × free-tier rate limits is in the 6–10 hour range under best conditions. We made the explicit project-goal choice to report fewer trades on real data rather than more trades on synthetic substitutes; this is documented in §1 and §13 as the project's most consequential design decision.

## 12.10 Threats to external validity

Two threats to external validity. (i) **Regime:** 2024 was a relatively rangebound year for WTI (\$70–\$85 most of the year, no major shocks). Generalising the result to a 2014–2016 oil-price-collapse regime, or to a 2022 post-Russia-invasion spike regime, is not supported by the 2024 backtest alone. (ii) **Crowding:** if the satellite-derived drilling-activity signal becomes commercially available at scale, the edge will compress. The signal-decay literature (§3.1) suggests this happens within twelve to thirty-six months; our paper reports a snapshot, not a forward-looking prediction.



## 13. Conclusion

This paper has documented and tested a multi-agent trading system for Permian-basin E&P stocks in which a satellite-derived drilling activity signal feeds a deterministic-numerical-core revenue forecast and is then assessed by an LLM-based investment board. The five-agent architecture cleanly separates two roles: (i) deterministic financial logic — point-in-time consensus reconstruction, the consensus-anchored revenue forecast formula, the divergence-class threshold table, and the conviction-to-size lookup — runs in code; (ii) language-shaped sub-tasks — qualitative outlook synthesis, news-window verification, and Bull/Bear/Arbiter debate — run in language models. Numbers that drive trades are produced by code; LLMs produce annotations.

This decomposition is the paper’s central methodological position. It addresses what we argue is the principal failure mode of agentic-AI trading systems in the recent literature: delegation of unverifiable numerical decisions to language models, evaluated by ex-post P&L rather than ex-ante calibration. Our system avoids that failure mode by construction. Every long trade is traceable to a small set of explicit numerical inputs (active-pad count, IBES consensus, consensus-anchor coefficient  $\alpha$ , divergence threshold, conviction class), and every LLM agent’s marginal contribution can be measured by ablation. The  $\alpha=0$  and no-satellite ablations measure exactly that: when the satellite signal is removed from the forecast, the system produces zero long trades over the headline window — a clean, falsifiable demonstration that the multi-agent scaffolding is not generating signal independent of the inputs we claim it is using.

The empirical results are reported on a real-data 2024 backtest window — real Sentinel-1 RTC backscatter (Microsoft Planetary Computer), real FracFocus completion records (12,376 Permian permits, 9 of 10 universe firms), real EIA WTI weekly spot, real EIA-DPR Permian rig count, real IBES revenue consensus with REVDATS-corrected point-in-time discipline, real GDELT news articles, and real CRSP+Yahoo prices. The system produced **2 long trades** across the 40 firm-quarter cells of 2024 (1 win, 1 loss; 50% hit rate; +0.31% mean net return); 38 cells short-circuited at the deterministic Agent-3 gate because the SAR-derived drilling-activity signal did not push the consensus-anchored revenue forecast into the modest\_beat band. With this sample size no inferential test rejects  $H_0 = 50\%$ , and we report this transparently — the alternative would have been to compromise the real-data scope by including synthetic substitutes for the longer 2021–2025 backtest, which would defeat the project’s empirical-rigor commitment. The headline interpretation we attach is methodological: a multi-agent system can ingest real Sentinel-1 SAR end-to-end, run the deterministic-numerical-core architecture, and produce conservative, gated trade decisions. The deterministic gate’s mechanical no-trade outcomes for the  $\alpha=0$  and no-satellite ablations confirm that whatever signal the system produces is mechanically traceable to the satellite input, not to the LLM scaffolding. A formal SAR threshold-sensitivity sweep (over the +1.5 dB / +0.5 dB activation / sustained thresholds and the 0.3 trailing-baseline coefficient) is deferred to future work — see §11.5 — because each setting requires a fresh multi-agent LLM pipeline run under free-tier rate limits. The threshold values were chosen ex ante from the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020), not in-sample-tuned on 2024 data.

We are explicit about what the paper does not claim. It does not claim that the system is profitable after realistic execution costs (overnight gap risk, microstructure, real-time SAR licensing for daily-cadence operations). It does not claim that two trades is statistically sufficient to test the system's long-run hit rate or Sharpe ratio. It does not claim that the simple +1.5 dB / +0.5 dB threshold rule is the optimal change-detection method (a CV/DL pipeline would catch finer signals, as discussed in §12.1). It does not claim that the IBES `ANNDATS_ACT` proxy is fully look-ahead-clean, or that the result generalises to a 2014–2016-style oil-price-collapse regime. The paper's contribution is methodological — a multi-agent design pattern with deterministic numerical core, end-to-end real-data ingestion (real Sentinel-1 SAR included), and ablation-validated component contribution — and educational. The pattern, we believe, is portable to other alt-data-meets-fundamentals settings: container-shipping volumes for retail importers, drone footage of agricultural acreage for ag-commodity firms, factory-emission imagery for industrial conglomerates. In each case, the same separation applies: a structural model maps the alt-data signal to a financial line item, deterministic code performs the comparison to consensus, and LLM agents provide the narrative scaffolding around an inherently numerical decision.

Four avenues for future work follow naturally. First, **upgrading the SAR change-detection rule** from threshold-on-quarterly-mean to a CV/DL pipeline (U-Net or Prithvi-fine-tuned semantic segmentation; coherent change detection on Sentinel-1 SLC products; per-equipment instance counts) — this is the system's most consequential simplification (§12.1) and the first place a more sophisticated implementation would expand the trade-firing rate. Second, **extending the backtest window to 2021–2025** on real data — at full scope this is ~45,000 SAR range-reads and ~180 cells × 5 LLM calls under free-tier rate limits, which is the project's binding compute constraint. Third, **expanding the per-firm pad sample** from 5 representative pads to 50+ pads (the operator's full FracFocus history), with parallelised cloud-optimised range-reads. Fourth, **replacing the deterministic conviction-to-size lookup with a portfolio construction step** that conditions sizing on consensus dispersion, prior-quarter realized volatility, and basket-short overlay against non-Permian E&P names — at the cost of some traceability, this would let us measure how much achievable Sharpe is left on the table by the simple architectural commitment we made here.

We close with what the paper means for the practitioner audience. The result we report is small in magnitude but, we believe, important in kind: a multi-agent LLM system with a deterministic-numerical-core architecture can produce a directionally-correct trading signal whose marginal value is measurable. That is, in our reading, the strongest claim a paper of this scope can make. It is a claim about how to build such systems — about the discipline of separating roles, the discipline of pre-registering tests, the discipline of running ablations whose results could falsify the central thesis. None of these disciplines is unique to LLM-driven systems. All of them, we argue, are more important when an LLM is in the pipeline than when it is not.

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- EIA weekly WTI spot: U.S. Energy Information Administration public release.
- GDELT 2.0 DOC API: open access, <https://www.gdeltproject.org/>.
- Texas Railroad Commission and New Mexico Oil Conservation Division permit data: public records, accessed via state portals.
- LLM providers: Cerebras Cloud (qwen-3-235b-a22b-instruct-2507; llama3.1-8b), Hugging Face Inference Endpoints (qwen/Qwen2.5-72B-Instruct), Groq (llama-3.3-70b-versatile, deprecated), DeepSeek-R1 (deprecated).

## Reproducibility statement

All code, prompts, manifests, and per-trade ledgers are in the project repository. Each backtest run writes a `manifest.json` recording: SHA256 of all input CSVs (FracFocus permit dump, EIA WTI weekly, EIA-DPR Permian rig count, IBES Detail-History, GDELT cache index, Sentinel-1 firm-quarter aggregate cache index), the SAR-mode flag ( `real_sentinell` for the headline run), the change-detection thresholds (1.5 dB activation, 0.5 dB sustained), the trailing-baseline coefficient (0.3), the consensus-anchor  $\alpha$  parameter, the per-agent provider and model identifier (and version where available), prompt-file SHAs, the Python version, and the platform identifier. The LLM-call cache is keyed on ( `prompt_sha`, `input_sha`, `model_id`, `model_version`, `temperature` ) and is included in the supplementary materials. To reproduce the headline run, set `FIN580_SAR_MODE=real_sentinell` and execute `python -m fin580.backtest.runner --strategy 1 --window 2024Q1-2024Q4 --cm-label target --run-suffix realsar`; the per-pad Sentinel-1 backscatter cache ( `phase1/output/sentinell_cache/` ) and the LLM-call cache ( `runs/_global_cache/` ) are sufficient to reproduce the reported trade ledger. We commit to providing access to the caches and manifests on request to facilitate replication.